FT-950 Remote

User & Technical Manual

Yaesu FT-950 web remote



Yaesu FT-950 web remote via the SCU-17 — CAT, WebRTC voice, FT8/FT4, CW skimmer — for a Raspberry Pi.

Version 1.3 · 2026-10-26

github.com/CQ-DJ0SH/ft950-remote

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1 Overview

FT-950 Remote is a web remote control for the Yaesu FT-950 HF/6 m transceiver. It runs on a Raspberry Pi next to the radio, talks to it directly through a Yaesu SCU-17 interface — CAT at 38400 Bd and USB audio — and serves the whole station as one HTTPS page. No hamlib, no desktop software: every control is mapped to the radio's documented CAT command and checked by reading the value back.

The focus is digital operation (FT8/FT4 with WSJT-X's own decoder) and remote voice over WebRTC, with a CW skimmer, a HackRF panadapter from a second Pi, logging to Wavelog and QRZ.com, and remote power through a Shelly relay.



2 Features

- VFO panel with an LCD modelled on the FT-950's display: FT-950 style analog meter, signal path as lit cells, mode lamps, DSP bars, seven-segment frequency, clarifier, VFO-B, PO/SWR/ALC/ID/AF bars on transmit.
- Title-bar switches: split, ANT 1/2, tuner and ATU tuning, CLAR/CLR, FLT · IPO · R.FLT · AGC, voice profile and FT8 profile.
- Tuning by mouse wheel or drag on a digit, frequency entry, band and mode keys, a frequency shuttle, CTCSS (ENC/TSQ), MON, NAR and a scaled TX power control.
- Push-to-talk in the browser with a watchdog, PTT-LOCK, roger beep, 1750 Hz tone call, MUTE and DROP; 32 memory keys (click recalls, hold 3 s stores).
- Two-way audio over WebRTC/Opus: RX/TX gain, mic AGC, compressor, pre-emphasis, mic echo test, RX recorder with WAV download.
- FT8/FT4 decoding with jt9 on three CPU cores, WSJT-X style sequencer with auto-answer, waterfall, DXCC and worked-before flags.
- HF band conditions under the waterfall from real VOACAP for your locator: 80-40, 30-20, 17-15, 12-10 and 6m in green / yellow / red, with a sun/moon switch for day and night.
- CW skimmer: every CW signal in the passband decoded at once; send through the radio's keyer with macros.
- HackRF waterfall relayed from TM-V71 Remote, following the amateur band of VFO A.
- Logging to Wavelog and QRZ.com; worked-before index from Wavelog.
- Radio power through a Shelly relay with auto power-off.
- Dark and light theme, 90 % panel zoom, installable as a PWA.

3 Hardware and radio setup

What you need

- Raspberry Pi with Debian 13 / Raspberry Pi OS. A Raspberry Pi 5 is strongly recommended: FT8/FT4 decoding with jt9 is the heaviest job, and the Pi 5 decodes a full band in a fraction of the Pi 4's time. A Pi 4 works, but needs 4–6 s per period at depth D2 instead of 1.5–2 s.
- Yaesu SCU-17 between Pi and radio (USB to the Pi; CAT and RTTY/DATA cables to the radio).
- Optional: a second Pi with TM-V71 Remote and a HackRF One (panadapter), a Shelly relay in the radio's 13.8 V line (remote power).

The SCU-17 on the Pi

The SCU-17 contains a CP2105 dual UART and a TI PCM2903B USB codec. The CP2105's "Enhanced" port (interface 0) carries CAT, the "Standard" port carries PTT/CW on RTS/DTR. The codec is the audio path to and from the radio's DATA jack. Use the stable /dev/serial/by-id path in Settings — it survives re-plugging.

```
/dev/serial/by-id/usb-Silicon_Labs_CP2105_..._if00-port0   CAT, 38400 Bd 8N1
/dev/serial/by-id/usb-Silicon_Labs_CP2105_..._if01-port0   PTT/CW (RTS/DTR)
hw:CARD=CODEC,DEV=0                                        audio in/out
```

Radio menus

- 026 CAT RATE = 38400 Bd (3), 027 CAT TOT = 10 ms (0).
- 031 TX TOT as a safety net for remote operation.
- 050 DATA TX GAIN — transmit level from the DATA jack (FT8, TUNE, voice). Adjust with TUNE until the power is reached and ALC just moves.
- 051 DATA OUTPUT LEVEL — receive level to the Pi (the codec has no capture control).
- 054/055 (DATA shift / VOX) as described in docs/FT8-DECODING.md.

Audio from the Pi reaches the radio on the DATA jack, which the FT-950 takes in the DATA (PKT) modes. Remote voice therefore runs in DATA-U / DATA-L — the VOICE PROFILE sets that up for the current band.

4 Installation

```
sudo apt install wsjtx voacapl    # jt9, ft8code, ft4code; VOACAP
git clone https://github.com/CQ-DJ0SH/ft950-remote /opt/ft950-remote
cd /opt/ft950-remote/backend
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
mkdir -p certs && cd certs && bash ../../deploy/gen-server-cert.sh /path/to/ca
sudo cp ../../deploy/systemd/ft950-remote.service /etc/systemd/system/
sudo systemctl enable --now ft950-remote
```

The service serves https://<pi>:8444/. The certificate is signed by a private root CA (shared with TM-V71 Remote) with a random serial and the Pi's own address; the CA can be downloaded in Settings ▸ General and on the power-off plate. Browsers allow the microphone only over HTTPS, so install the CA once on every device.

Then open Settings ▸ General: callsign, locator, CAT port and baud rate. Station data, logbook credentials and memory names are stored in backend/runtime.json (mode 0600) and never sent back to the browser.

5 The page



The title bar shows the logo (uploaded under Settings, never part of the repository), your callsign (shown with a slashed zero), the nameplate, the CAT and USB AUDIO lamps, the UTC clock, the theme switch, Settings and — once a Shelly is set up — the power switch.

Below follow the panels VFO, PTT · MEMORY and AUDIO, DIGI, CW, HackRF and LOG. Most can be collapsed with ▲; PTT and AUDIO always stay open. The panels are shown at 90 % zoom so the whole station fits on one screen.

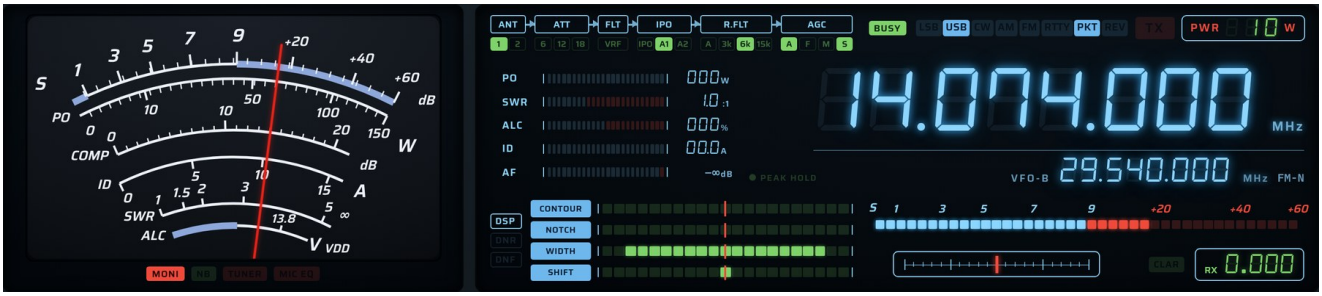
6 VFO panel



Title bar

- LOCK and PLL lamps: dial lock, PLL unlock (a radio fault). A click on LOCK (or Enter/Space) locks or unlocks the dial (LK, verified by read-back).
- SPLIT: receive on VFO A, transmit on VFO B.
- ANT 1 / ANT 2, TUNER (internal ATU in/out), ATU TUNE (starts a tuning cycle — the radio transmits a carrier; confirmation required).
- CLAR switches the RX clarifier, CLR sets its offset to 0.
- FLT · IPO · R.FLT · AGC step through VRF/THRU, IPO/AMP1/AMP2, AUTO/15k/6k/3k and OFF/FAST/MID/SLOW/AUTO; the LCD shows the state.
- VOICE PROFILE and FT8 PROFILE: read every value, write only what differs, verify by read-back, report what changed.

LCD



Left window: the analog meter — S on receive, PO on transmit — with the FT-950's scales (S, PO 0–150 W, COMP 0–20 dB, ID, SWR 1–∞, ALC/VDD 13.8 V), white on black with steel-blue bands, and the MONI/NB/TUNER/MIC EQ indicators.

Right window: the signal path ANT · ATT · FLT · IPO · R.FLT · AGC as fixed cells (only the active values light up), BUSY, the mode lamps LSB USB CW AM FM RTTY PKT REV (DATA-U lights PKT + USB, CW-R lights CW + REV), TX lamp and set power, the PO/SWR/ALC/ID/AF

bars, the DSP bars, the frequency to 1 Hz, the clarifier scale (red mark = offset, ±1 kHz, a tick every 100 Hz) with its value, VFO-B under VFO A and a digital S meter.

The digital S meter under VFO-B is a 40-segment bar, blue to S9 and red above, with the scale 1 … 9, +20, +40, +60 at the needle's S points; it stays dark while transmitting. As on the FT-950's LCD, every 7-segment digit place shows a dimmed 8 and unlit bar segments are dimmed, and nothing changes colour — the PWR box keeps its colours on transmit, the TX lamp shows it.

PO and SWR use approximate calibration points; ID (amps) follows the meter's 5/10/15 A marks; AF shows the microphone level on voice and the tone drive on FT8/TUNE (computed from FT8 TX DRIVE, not measured). Peak hold: while transmitting a held segment marks each meter's highest reading; after the transmission the bars and values show these peaks — the values in a green box, PEAK HOLD lit beside it — until the next TX starts (switchable in Settings ▸ Appearance). SWR counts only with some power out. Values keep a fixed number of digits (SWR as 1.5:1), so nothing jumps. The needle moves like a moving-coil meter: damped, rising in about 0.35 s and falling back a little slower.

Tuning

- Mouse wheel or drag on a digit of the LCD frequency.
- Frequency entry with ▲▼ (mic UP/DN), A⇄B and A→B.
- BAND and MODE keys light up at once and confirm when the radio reports back.
- Shuttle next to BAND (◀ DOWN · TUNE · UP ▶): hold left of the centre to tune down, right to tune up; the further out, the bigger the steps (10 Hz, 100 Hz, 1 kHz, 10 kHz) — the label shows the current step.
- CTCSS: off / ENC (send the tone) / TSQ (tone squelch) and the 50 tones, generated by the radio. MON switches the monitor. NAR selects the narrow filters.

TX power and RX · FILTER · LEVEL

The red TX POWER control sets 5–100 W. RX · FILTER · LEVEL opens the groups RX (preamp, AGC, ATT), FILTER (width, shift), LEVEL (RF gain), CLAR and DSP. The roofing filter is switched with R.FLT in the title bar.

- CLAR: ON switches the RX clarifier, CLR sets the offset to 0; the slider sets the offset (±2.5 kHz in 10 Hz steps, double-click = 0) and switches RX CLAR on. The radio's offset is read back and checked.
- DSP: DNF, NR and NB as switches.

7 PTT · Memory



- PTT: hold the button or the space bar. The browser renews a lease every second — if the page goes silent, the Pi drops PTT after 2.5 s. Maximum 180 s per over. The microphone is muted unless you transmit.
- PTT-LOCK latches transmit; ROGER adds a roger beep; 1.750 kHz arms the repeater tone call: the next PTT sends 1750 Hz instead of the microphone, releasing PTT disarms it (FM modes).
- MUTE mutes the receive audio (descending chime), DROP blanks only the current over and lifts when the station stops (ascending chime).
- PTT → VFO shows which VFO transmits. In the installed PWA, level bars for RX and MIC stand left and right of the PTT button.
- Memory keys M1–M32 (2 × 8 per side): click recalls frequency and mode to VFO A, in FM with CTCSS, plus VFO B and split if the key has them; hold 3 s to store VFO A in that channel (confirmation before overwriting).

Settings ▸ Memories edits all 32 channels: frequency, mode and CTCSS (mode and tone) go into the radio with MW, verified by MR; split, VFO B and a name are kept on the Pi, because the FT-950's memories have none of them. DEFAULTS (M1-M32) fills one band per key row for 80/40/20/15/10/6 m — left SSB | CW calling frequencies, right FT8 | FT4 — then 10/6 m FM, the 10 m repeater (split −100 kHz), SSTV and the IARU Region 1 emergency centres.

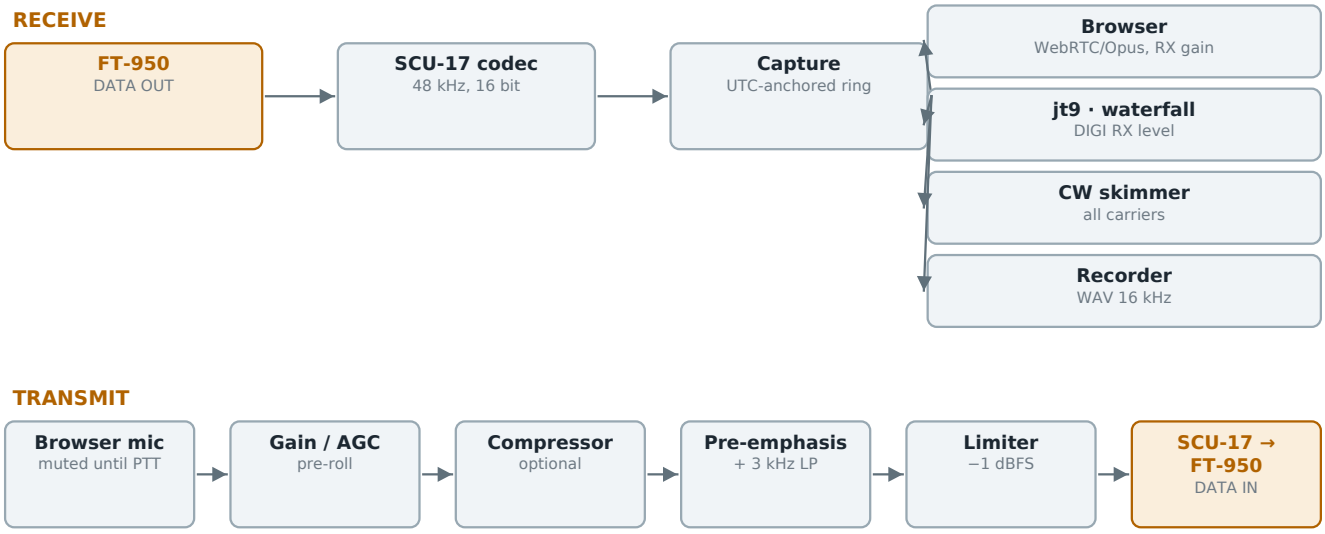
8 Audio



CONNECT starts WebRTC: the radio's audio plays in the browser, the microphone goes to the radio while you transmit. RX GAIN and TX GAIN set the levels. The switches in the title row: AGC rides the microphone toward a target level; MIC TEST records your microphone through the whole path and plays it back without transmitting — at listening level, the meters keep showing the real transmit level. On a desktop the browser's own microphone processing stays on (switching it off drops the level by about 40 dB on macOS); a built-in microphone is preferred to a Bluetooth headset.

REC records the receive audio on the Pi (up to 30 minutes, 16 kHz), ▶ plays it, ↓ WAV downloads it. Settings ▸ Audio adds a compressor and pre-emphasis for SSB voice, the FT8 TX drive, the codec output level and the FT-950's audio menus with saved defaults.

The graph is split: on top the levels of the last 30 s, below the audio spectrum 0–3 kHz (the FT-950's SSB/DATA passband) — receive audio in violet with a yellow peak hold that falls away in about 10 s, your microphone in red while you transmit or run MIC TEST. PEAK in the legend switches the peak hold. The spectrum is computed in the browser and starts with CONNECT.



FT8/FT4 and TUNE: tones generated on the Pi (ft8code/ft4code) go straight to the codec; CW is keyed by the radio itself (keyer memory).

9 DIGI — FT8 / FT4



DECODE in the panel's title bar switches decoding on and off (as in the CW skimmer). Decoding runs every period with WSJT-X's jt9 — or a faster build chosen in Settings ▸ FT8 ▸ Decoder (see below); the WSJT-X plate in the title bar shows the one that runs (2.7, 2.7 fast, 3.3 MT). The passband is split over three cores. D1/D2/D3/AUTO in the strip pick the depth: D2 (default) puts the decodes on screen about 1.5–2 s after the period on a Pi 5, D3 finds a few more weak signals in about 3 s, AUTO follows the measured run time. Received messages list SNR, time offset, frequency, message and DXCC, with worked-before colouring.

- Click a decode to put the RX window there (AF — adopt frequency — also moves TX; Auto Answer and Auto Hunt then answer on the partner's frequency, and the TX box follows while their QSO runs); double-click answers the station — in Received and in My Messages. A double-click within the first ~1.7 s of your own TX period still transmits in that period (start at once, the partner sees a slightly positive DT); later it goes out in the next one.
- The filters above Received Messages show all, CQ only, RR73 (with RRR) only or 73 only. follow keeps the newest row of each list in view, Session Messages included.
- A failed session shows just error in the list; the reason comes as a toast (when it happens and on a click on the row) and as the tooltip.
- With a session marked (click its row), QRZ Visit opens the station's QRZ.com page and Map Display a world map: position from the locator (without one, the centre of the DXCC entity), great circle, distance and bearing from your locator. Both open a popup window.
- CQ calls CQ; Auto Answer: while idle, a station calling you is answered automatically.
- Auto Hunt calls new stations by itself: while no QSO runs, after every full decode pass it

calls the best new CQ (a plain CQ, not CQ DX/NA/…; new DXCC first, then the strongest) and waits for the QSO to end. A station that never answers is struck through in Received Messages and not called again (the list survives restarts). While on, the button is red and shows Hunt and the time left as m:ss (∞: no switch-off). It calls only while a browser shows the DIGI panel. Whom it calls, the minimum SNR, the switch-off time (default 180 min, 0 = never) and the no-reply list are in Settings ▸ FT8.

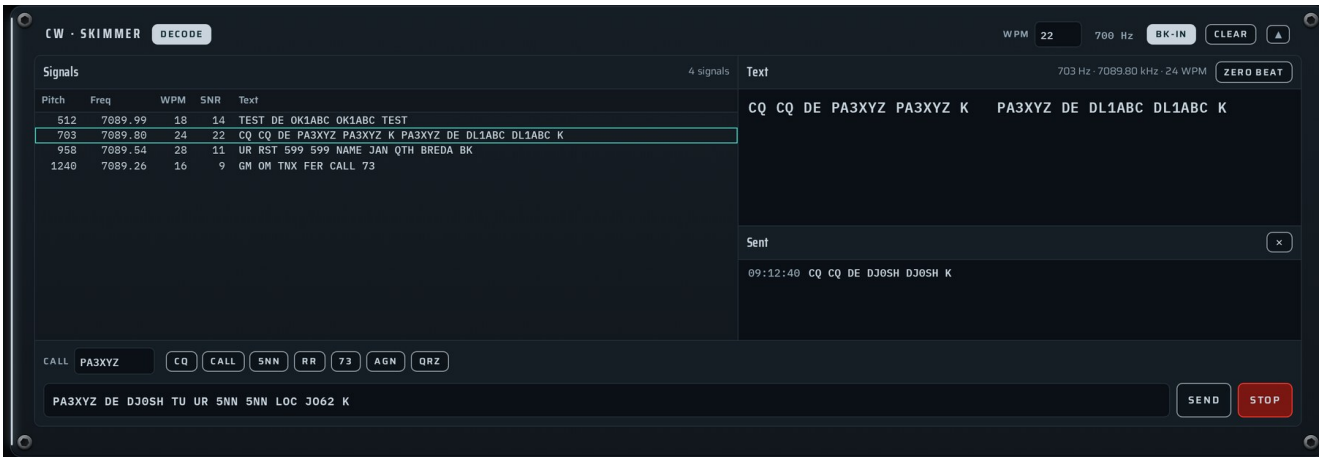
- The sequencer follows WSJT-X: call, report, R-report, RR73, 73; it can log automatically: Auto-log in the Sessions header (default on) logs every completed QSO to Wavelog (the QRZ.com logbook by hand, LOG QRZ.COM), otherwise it is offered in the log form. A reply that comes in while our older message is still on the air is answered next period — the QSO is not closed before our own 73. While a QSO runs, a narrow decode around the partner (FT8 ±200 Hz, FT4 ±300 Hz — FT4 needs the wider margin) has their reply within ~0.1 s of the period end, so the next message usually starts 0.5–1 s into our period; without a reply there it repeats at once. It can send RRR instead of RR73 (Settings ▸ FT8).
- The slider with the speaker sets the DIGI RX level on the Pi (the meter for it, target −40 … −20 dBFS, is in the footer next to the dial frequency), the one with the antenna next to it the FT8/FT4 TX drive (as FT8 TX DRIVE in Settings ▸ Audio); TUNE sends a steady tone. Halt TRX, right in the title bar, is active during an FT8/FT4 transmission, a sequence or while Auto Hunt is on (dimmed otherwise); it stops at once and switches Auto Hunt off. NTP in the strip is an LED: green with the clock synchronised, red without (no decoding).

Band conditions (VOACAP)

At the bottom right of the panel a sun or a moon and five chips — 80-40, 30-20, 17-15, 12-10, 6m — show how the bands are open from your locator, in green, yellow or red, as iDigi shows them. The values are computed on the Pi with VOACAP (voacapl): VOACAP predicts a path, not a place, so the question goes to a ring of 36 paths around you (12 directions × 1500 / 4000 / 9000 km) at 100 W FT8 with an isotropic antenna. Per band the reliability is the mean over the ring, and a pair is as good as its better band: green from 0.6, yellow from 0.3, red below. Hovering a chip shows both bands in per cent and the trend of the next three hours.

- Sun or moon follows the real time of day at your locator. A click shows the other half of the day — the mean over today's daylight or darkness hours there — with a framed switch and dashed chips; a second click goes back to now.
- Updates: one run covers all 24 UTC hours of the month's median ionosphere (about 0.5 s, at low priority so jt9 never waits). It is computed once per UTC day and again when the locator or the sunspot number changes. The sunspot number is NOAA SWPC's latest monthly value, fetched once a day; the last good one is kept for an offline Pi. The page asks every 5 minutes, so the next hour, sunrise and sunset show within 5 minutes.
- Limits: VOACAP is a monthly median of the F2 layer. It does not see flares or geomagnetic storms, and on 6 m it shows only the rare F2 openings of a solar maximum — not sporadic-E, which opens 6 m in a European summer. A red 6m chip does not rule out Es.
- Without voacapl, or without a locator in Settings ▸ General, the strip is not shown.

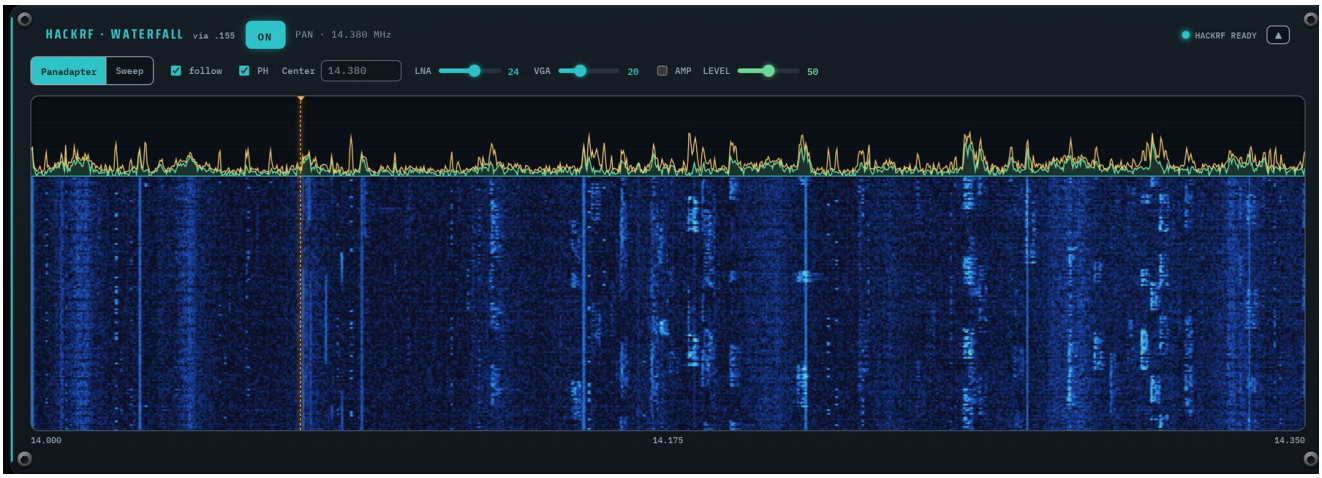
10 CW skimmer



DECODE starts the skimmer. Every CW carrier in the passband gets its own decoder; the table shows pitch, RF frequency, speed, SNR and the running text. Click a row to follow it: its text appears under Text, ZERO BEAT tunes VFO A onto it.

Sending uses the radio's own keyer: the text is written to keyer memory 5 and played (KM/KY), in chunks of 50 characters. The radio must be in CW or CW-R with BK-IN on. Macros (CQ, CALL, 5NN, RR, 73, AGN, QRZ) fill the send line; CALL is taken from the selected signal.

11 HackRF waterfall



The HackRF One sits on the second Pi with TM-V71 Remote. FT-950 Remote relays its waterfall (Settings ▸ General ▸ HackRF source). With follow on, the view shows the whole amateur band of VFO A; one click tunes VFO A there. The HackRF is shared: what you set here also applies to the TM-V71's waterfall.

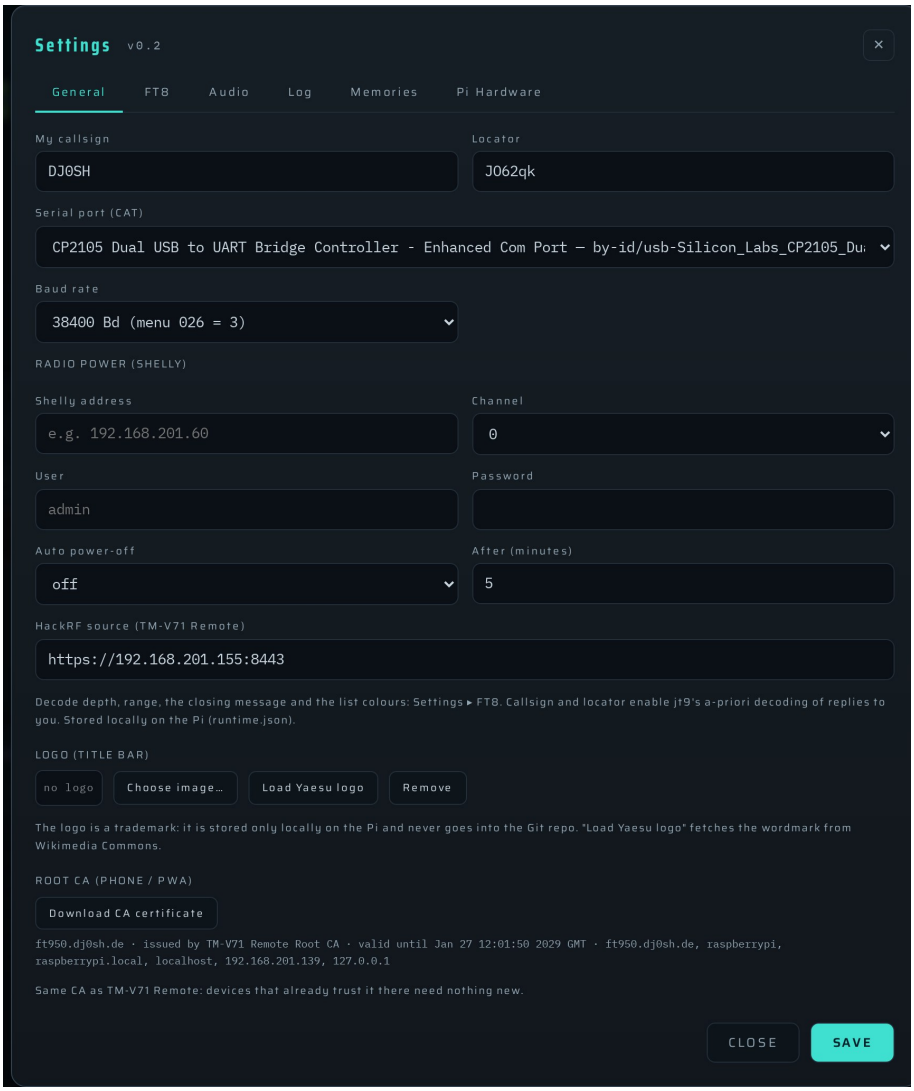
In the band view TM-V71 Remote sends a fine row of just that band (8192-point FFT, 244 Hz per point, 4 blocks averaged) — 820 points for 40 m instead of about 50 from the normal 2 MHz row. The mouse shows frequency and level under the pointer.

12 Log

The LOG panel logs to Wavelog (API key and station profile) and the QRZ.com logbook — each with its own button beside the callsign: LOG WAVELOG and LOG QRZ.COM (the form stays, so a QSO can go to both; a ✓ marks where it went). QRZ LOOKUP fills name, QTH and locator; NEW STATION / WORKED shows in the title bar. The worked-before index is downloaded from Wavelog and colours the FT8 list. Settings ▸ Log holds the accounts — the keys never leave the Pi.

In the title bar, LEDs as in the page's own title bar: WAVELOG and QRZ green when set up, AUTO green while completed QSOs are logged automatically (to Wavelog). The LOG buttons are only outlined while the form is empty and filled once a callsign is in it — then there is something to log. Recent QSOs: QRZ Visit and Map Display for the marked row.

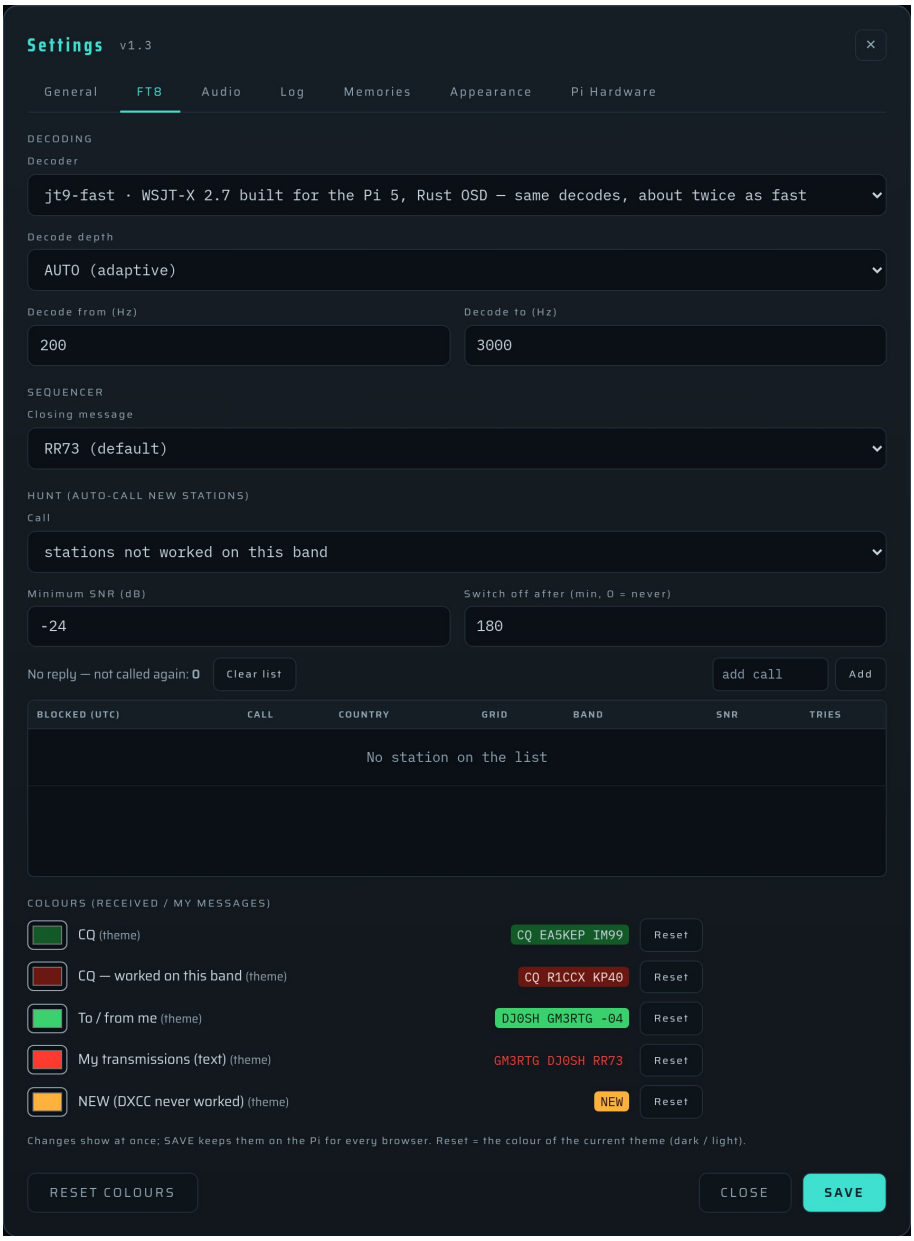
13 Settings

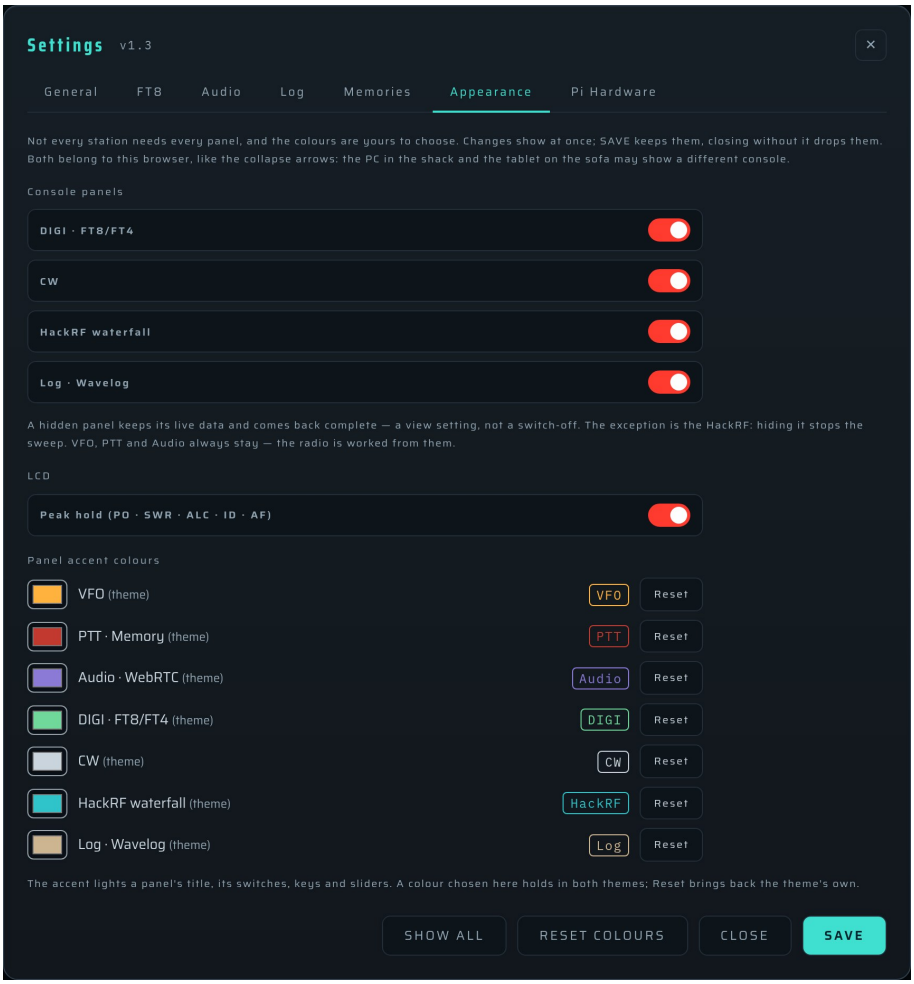


- General: callsign, locator, CAT port and baud rate, Shelly and auto power-off, HackRF source, logo, root CA.
- FT8: the decoder — jt9 (WSJT-X 2.7, Debian package), jt9-fast (the same decoder built for the Pi 5 with its OSD stage in Rust: identical decodes, about twice as fast) or jt9-33 (WSJT-X 3.3 beta, multithreaded MTft8 with a Rust OSD: about 6 % more decodes, ready about 2.4 s after the period; it sets its own depth). Builds not installed are greyed out. Then decode depth (as D1/D2/D3/AUTO in the panel) and range, closing message RR73 or RRR, Auto Hunt (whom to call: not worked on this band, never worked, DXCC new on this band, DXCC never worked; minimum SNR; switch-off time, 0 = never; the no-reply list as a table — blocked time, call, country, grid, band, SNR, tries — with fixed column heads, × to take a call off, a field to add one by hand and Clear list; these edits apply at once), and the colours of the message lists with preview and reset.
- Audio: capture device, codec level, FT8 TX drive, the FT-950 audio menus, voice TX processing, WebRTC buffers.
- Log: Wavelog and QRZ.com.
- Memories: the 32 memory keys with CTCSS, split and VFO B.
- Appearance (per browser): hide the DIGI, CW, HackRF and Log panels (VFO, PTT and

Audio always stay; hiding the HackRF stops its sweep), an accent colour per panel with Reset, and peak hold of the LCD meters on/off. Changes show at once and are kept with SAVE.

- Pi Hardware: load, temperature, memory, disk and a 24 h graph.





Every tab ends with the same bar: CLOSE, plus SAVE on General, FT8, Log and Appearance. On Audio changes apply at once, on Memories each row has its own WRITE, Pi Hardware is read only.

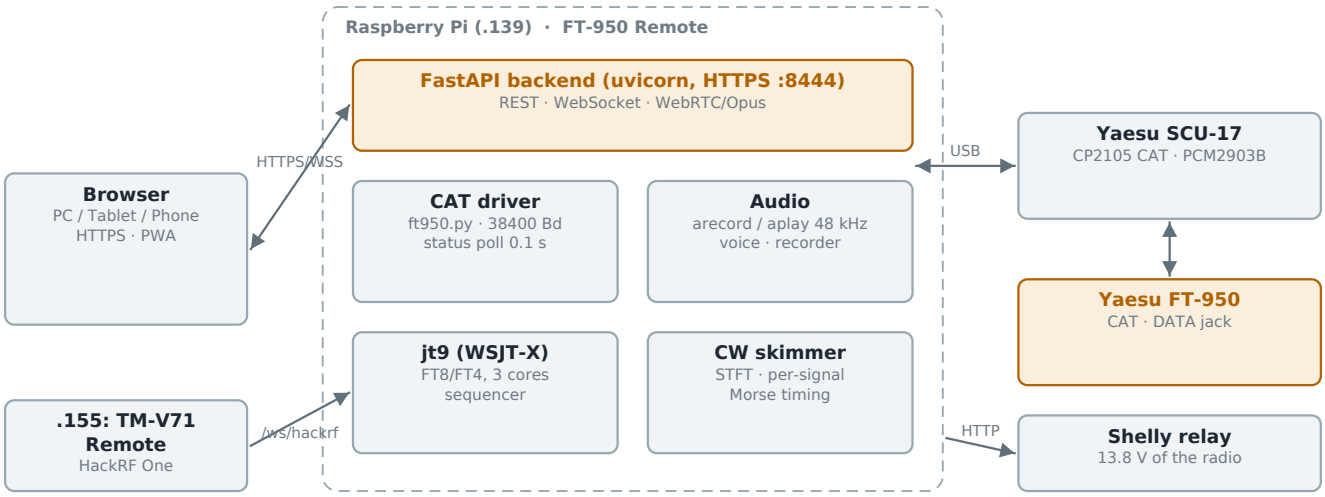
14 Remote power

A Shelly relay (Gen1 or Gen2/Plus/Pro, detected automatically) switches the radio's 13.8 V. The power switch in the title bar turns it on and off — before cutting the supply, a transmission and the sequencer are stopped. After switching off, power stays off for at least 3 s. While the radio is off a plate covers the page. Auto power-off switches the radio off when no browser has had the page open for the set time.

15 Troubleshooting

- CAT offline: check the by-id port and menu 026 = 3 (38400 Bd).
- No receive audio: CONNECT, then check RX GAIN and that MUTE/DROP are off.
- Nothing decoded: NTP must be synchronised (NTP lamp), mode DATA-U, the DIGI level in the green range.
- FT8 too quiet: FT8 TX drive (Settings ▸ Audio) and menu 050.
- No microphone: the page must be opened over HTTPS with the trusted CA.
- After a backend restart: reload the page and CONNECT again.
- No band strip under the waterfall: apt install voacapl, and set a locator in Settings ▸ General.

16 Technical reference



Backend

- FastAPI on uvicorn, systemd unit with RuntimeDirectory, StateDirectory, memory limits and Restart=always.
- ft950.py: CAT driver with read-back verification; radio_service.py polls frequency/mode/meters every 0.1 s and the receive chain every second.
- digi.py: UTC-anchored slices, jt9 in parallel sub-bands, adaptive depth, early and partner passes; sequencer.py: the QSO state machine.
- voice.py: WebRTC hub, PTT lease, pre-roll, AGC, compressor, pre-emphasis.
- cw.py: the CW skimmer; hackrf_proxy.py; shelly.py; recorder.py.
- propagation.py: VOACAP band conditions — the ring of 36 paths, one run per UTC day, the day/night halves; GET /api/propagation.

Files on the Pi

```
backend/runtime.json      station data and credentials (0600, not in git)
backend/ex_defaults.json  saved FT-950 audio-menu defaults
backend/certs/            server certificate (from the private CA)
/run/ft950                decoder slices (tmpfs)
/var/lib/ft950            jt9 working data
backend/itshfbc/          VOACAP working tree (makeitshfbc copy)
backend/propagation.json  last sunspot number from NOAA
```

Source, issues and updates: github.com/CQ-DJ0SH/ft950-remote — GPL v3.